ABCXM: A BENCHMARK FOR MULTIMODAL MODELS PLAYING SCORES TO MUSIC

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ABSTRACT

We introduce a new task of converting scores to music, addressing a gap in the capabilities of current multimodal models. Unlike traditional Optical Music Recognition (OMR) methods, our composite pipeline converts music scores directly into music (audio). We created the ABCxM benchmark, which includes four datasets in ABC, MIDI, and WAV formats: "ABC4M_120K," "ABC8M_120K," "ABC16M_90K," and "ABCfull_5K." Carefully selecting embedding features based on human perception via Amazon Mechanical Turk, we chose Fréchet Audio Distance and Dynamic Time Warping scores for both data and sample-level evaluation. Fine-tuned on our datasets, our best baseline model achieves a 0.82 FAD score on MERT-v1-95M embeddings. Our experiments highlight the varying performance based on three knowledge incorporation strategies and the challenges posed by different difficulty levels, providing insights for future work.

***Index Terms*—** benchmark, music evaluation, music scores, FAD score, music embedding

1. INTRODUCTION

This paper introduces an end-to-end benchmark designed to evaluate models that directly convert music scores to audio. Traditional Optical Music Recognition (OMR) models focus on classifying or detecting music components such as notes, measures, and staves [1]. Yet our task examines models' capabilities of reading graphic scores and evaluates auditory outputs directly.

Given the success of modern any-to-any models, it is motivated by the following reasons. 1) Not everyone understands OMR recognition results, as it requires expertise. Tools like MuseScore and Finale are mainly designed for professionals. For general use, such as roughly knowing how a picture of a score sounds, these tools are too specialized and lack accessible benchmarks; 2) Many current OMR evaluation methods rely on object-level comparisons using Average Precision (AP), which do not adequately represent music interpretation. We propose a wave-level comparison; 3) The task of converting notation to music remains underexplored. Within the scope of our search, this Notation-to-Music has not been conquered by today's multimodality models.

Our major contributions are as follows: First, we compose large-scale, multi-length music score datasets paired with textual notation files and audio files as well. Second, we fine-tuned nine models on our training splits with customized prompting strategies. Our experiments show that incorporating a certain level of music knowledge complexity during training and testing improves model performance, but overwhelming the model with excessive knowledge decreases its performance. Third, we carefully compared different embeddings as the basis of our evaluation metrics. Based on human annotations through Amazon Mechanical Turk (AMT) [2], we selected MERT-v1-95M for our basic data and sample-level evaluation embeddings. Fourth, through comparisons among similarities in textual output, auditory music, and human annotations, our results show that auditory level evaluation is closer to human judgment than textual level comparison, underscoring the significance of this task.

2. BACKGROUND AND RELATED WORKS

2.1. Datasets from Optical Music Recognition

Optical Music Recognition (OMR) develops computational methods for reading and interpreting music notation. Unlike Optical Character Recognition (OCR), OMR lacks a clear definition due to its specialized nature, attracting fewer researchers [3]. Over its 50-year history in document image analysis, recent OMR advancements are driven by deep learning models, with most research focused on parsing score components using object detection frameworks [4, 5].

Many score datasets come from OMR collections. The Handwritten Online Musical Symbols (HOMUS) dataset [6], with 15,200 symbols, is used for both online and offline symbol classification. The CVC-MUSCIMA dataset [7] consists of 1,000 handwritten score images for staff removal and writer identification, while its extension, MUSCIMA++ [8], contains over 90,000 annotations for symbol detection and end-to-end recognition. The DeepScores V1 and V2 datasets [9, 10], containing hundreds of thousands of typeset images, support object detection and semantic segmentation tasks. A recent dataset named MusicScore is limited to metadata and

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full-length score images only [11]. However, these datasets lack textual notation or audio files that can support our proposed task.

The PrIMuS dataset comprises 87,678 incipits in formats of images and MEI, making it suitable for end-to-end conversion [12]. Similarly, the DoReMi dataset contains approximately 6,432 images of sheet music featuring nearly one million annotated objects [13]. Both datasets primarily consist of relatively short musical scores, with most pieces containing only a single line of music. Additionally, due to its excessive token length, the MEI format is not particularly suitable for large language models (LLMs).

2.2. Textual Encoding Format

Many studies use MusicXML, a widely accepted format for digitizing scores [14, 15]. Due to the challenges in converting music scores to a linear sequence, derived formats like Linearized MusicXML [16] have been proposed. Another popular format is MEI [17], which is similar to MusicXML but better suited for digital archives and provides a more detailed encoding system. Some machine learning research chose MEI [18, 19].

However, for language models, these formats are too lengthy in terms of token limitation. Researchers choose shorter text-based syntaxes, such as **kern and ABC. The former is part of the Humdrum toolkit for computational music analysis [20] and is a popular choice for music encoding in computational research [21, 22]. The ABC format is another plain-text format [23] and is easier for humans to read and understand compared to **kern. Its simplicity and conciseness make it ideal for automated music composition and generation [24, 25].

2.3. Parsing Notation with SOTA Models

Any-to-any models that convert images to music do exist. For instance, AnyGPT can generate music by interpreting prompts like “give me music whose style is close to this image” through interleaved dialogue [26]. CoDi can generate sound from reasoning and time-sequenced pictures [27]. Nevertheless, none of these models can accurately parse and interpret music notation.

Advanced models like GPT-4, Gemini, and Claude seem capable of parsing score images into textual notation and converting them into MIDI files. However, a simple test using a 4-measure score image shows that these models only produce ABC note samples without accurately recognizing the notes. Some studies explore fine-tuning language models to understand music notation, but they often use smaller models like GPT-2 or focus on music information retrieval [28, 29].

Considering the poor performances of current models, we planned to pipeline our baseline into two steps: 1) fine-tuning existing LLMs to parts score images and 2) converting the outputs into music files.

3. EXPERIMENTAL SETUP

3.1. Benchmark Dataset

To address the inadequacy of existing datasets for our task, we generated a series of music score images with varying measure lengths. We downloaded 105,278 ABC format pieces from the ABCNotation website and processed them into three datasets of 4-measure, 8-measure, and 16-measure segments, with overlaps of 1, 3, and 6 measures, respectively. These segments were converted into JPG, MIDI, and WAV files.

Subsequently, we landed on four benchmark datasets: “ABC4M_120K” contains 122,754 4-measure notations, “ABC8M_120K” contains 121,131 8-measure notations, “ABC16M_90K” contains 90,045 16-measure notations, and “ABCfull_5K” contains 4,994 full-length notations. The first three were split into training, validation, and test sets, and the last one was only a test set. (Fig 1)

Fig. 1. Examples of score images from ABC4M, ABC8M, ABC16M, and , ABC

3.2. Baseline Models

As Section 2 mentioned, our plan for the baseline model is decided as “score image - notation - music.” We chose ABC format as our textual media [23] since its legibility for humans and conciseness compared to MusicXML and MEI. This is consistent with language models’ token limitation. For example, an “Ode to Joy” has 282 tokens in ABC format but takes 5,179 tokens in the form of MusicXML. [1]

Our baseline model starts from LLaVA 1.5-7B due to its clear repository and ease of use [30]. The training data was composed in a question-answer pair formatted in JSON as required by LLaVA. The question is fixed in each pair as “Read this notation and write it out in an ABC format,” either followed by a piece of textual knowledge of ABC format or not. In our experiment, we tried three prompt combinations: ”question alone”, “question + less knowledge”, and “question + more knowledge”. We fine-tuned 9 models on the ABC4M, ABC8M, and ABC16M training splits with these three strategies through the LoRA script with the default settings and

[1] the token counts are based on https://platform.openai.com/tokenizer

configurations. Each model took approximately 20 hours to finish one epoch on an 8-GPU AWS G5 instance. Every fine-tuned model was evaluated over the four test sets.

3.3. Evaluation Metrics

Original and Revised output After fine-tuning, our models can produce ABC files when given an image of notation. However, many textual outputs missed the "header part," which defines various metadata such as title, meter, tempo, key, default note length, etc., thus causing failure to be parsed as a music file. Accordingly, we relaxed our evaluation for two versions of outputs – an original prediction and a revised prediction, which only contains the body part from the output but is attached with the header transplanted from the ground truth.

We kept track of the percentage of ABC files that can be successfully converted to WAV files as the conversion rate. The conversion rates for the original ABC outputs were less than 25%, yet that of the revised notes increased up to 99%.

Textual Evaluation. Average Normalized Levenshtein Similarity (ANLS) was chosen for the text-level output comparison. The scores were calculated separately for the header, body, and the combined header and body, comparing predictions with the ground truth.

Auditory Evaluation (Data-Level) For data-level comparison, Fréchet Audio Distance (FAD), which is given by the equation below, has been extensively used in audio generation evaluation [31]. The FAD measures the difference between the mean (μ_r, μ_g) and covariance ($\mathbf{\Sigma_r}$, $\mathbf{\Sigma_g}$) of features extracted from real and generated audio distributions, thereby quantifying the quality of generated audio about real audio samples. We selected the FAD tool kit for this evaluation because it provides a series of embeddings from which to choose [32]. Since our task focuses solely on music generation, KL divergence, commonly used to measure output diversity across different audio classes, is not suitable for this context.

$$\text{FAD} = \|\mu_r - \mu_g\|^2 + \text{tr}(\mathbf{\Sigma_r} + \mathbf{\Sigma_g} - 2\sqrt{\mathbf{\Sigma_r}\mathbf{\Sigma_g}})$$

Auditory Evaluation (Sample-Level) For sample-level dataset comparison, we calculated Dynamic Time Warping (DTW) [33], which measures the alignment between generated music and the ground truth when their lengths differ. DTW computes the optimal alignment between two sequences by minimizing the cumulative distance between corresponding points, allowing for temporal distortions.

Human annotation of Music Similarity The key to FAD and DTW lies in feature extraction since scores calculated from different embeddings are not comparable. We conducted an AMT survey to collect auditory similarity ratings from humans. Each survey question contained a pair of ground truth and predicted music pieces, asking participants whether the two pieces sounded similar. Turks rated the similarity on a scale from 1 (almost identical) to 5 (not similar). We prepared 500 pairs of 4-measure length music, and each piece lasted about 10 seconds, which made it relatively easy to remember the previous tune after listening to the second piece and point out differences if they existed for Turks.

4. EXPERIMENTAL ANALYSIS

4.1. Alignment with Human Annotation

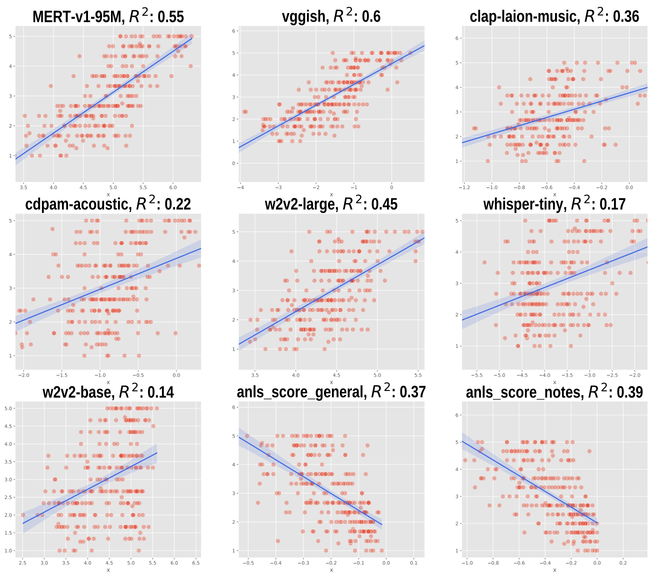

Fig. 2. R^2 of linear models with different embeddings

We gathered 1,542 hits, with each question rated by at least three participants. Responses from 51 qualified participants were accepted, resulting in 66 out of 500 pairs receiving identical answers across all three ratings. We utilized the similarity level as an ordinal value, allowing the standard deviation for each question to indicate the degree of discrepancy in the answers. Notably, 72.4% (362) of the questions exhibited a standard deviation below 1, indicating relatively high agreement.

Based on these results, we employed linear regression models using various embeddings with DTW distance to predict average similarity scores for questions with low standard deviations, indicating human consensus. After selecting cosine distance as the metric and applying a standard deviation filter of 0.8 for similarity, we evaluated a series of embeddings, as illustrated in Fig 2.

Among all embeddings, VGGish achieved the highest R^2 value. However, upon examining the domains of the embeddings, it becomes evident that MERT-v1-95M and clap-laion-music are specifically related to music, while others are designed for general sound or human voice classification. VGGish, in particular, is more suited for sound classification.

Subsequent tests using VGGish to compute FAD further supported this: all FAD scores based on VGGish were quite low, with most scores falling below 1. This is likely because when comparing the similarity between two music pieces using VGGish, both pieces are predominantly classified as "music" or "piano music," leading to extremely low FAD scores that diminish the metric's evaluative significance. Therefore, we selected MERT-v1-95M, and subsequent tests confirmed it as an appropriate embedding that more accurately reflects model performance.

Additionally, we attempted to regress the ANLS score on human-annotated similarity values, but the R^2 values were lower than those obtained using waveform methods. This further supports the notion that direct music-to-music comparison is more effective than text-to-text comparison.

4.2. Baseline model results

The performances of our models on all the test sets are shown in Table 1 using the FAD score based on MERT-v1-95M embeddings. Models trained on a specific type of notation perform best on the same type of test sets. When detailed ABC knowledge is incorporated during training or testing, the evaluation results are worse than those with less detailed knowledge. This may be due to the token size limitation of LLaVA 1.5-7B, a relatively lightweight model that struggles with large token knowledge. Models trained without additional knowledge have higher FAD scores than those incorporating knowledge during training and testing. This pattern is consistent across models trained on datasets of different lengths. The best model in our experiment was trained on 8 measures using less-detailed ABC knowledge, achieving an average FAD score of 8.52 across all four test sets. FAD scores for full-size notation images are quite high, indicating sensitivity to the scale of notation. Achieving better performance requires a robust model with comprehensive strategies. (Table 1)

Table 1. FAD score based on MERT-v1-95M embeddings

ID	Train Stage		Test Stage				
	Data	Knwl	Knwl	4M	8M	16M	Full
1	4M	—	Less	2.11	8.48	14.06	24.80
2			More	12.97	11.73	21.82	25.11
3		Less	Less	1.47	4.79	15.46	18.33
4		More	More	23.88	31.25	36.94	58.42
5	8M	—	Less	2.49	1.66	7.14	24.79
6			More	26.56	21.63	62.89	47.43
7		Less	Less	5.11	0.82	4.36	23.79
8		More	More	13.8	16.28	15.80	20.78
9	16M	—	Less	6.96	3.93	5.36	28.90
10			More	21.59	29.55	18.27	41.28
11		Less	Less	7.52	3.87	3.60	19.85
12		More	More	29.19	33.28	31.94	18.15

Notes: **None** means no ABC knowledge, **Less** means limited, and **More** means detailed ABC knowledge.

5. CONCLUSIONS

In this research, we introduced a pure end-to-end task: Notation to Music. Unlike traditional OMR studies, this task emphasizes directly evaluating the quality of audio files generated from music scores. We utilized AMT for human annotations to assess both textual and audio similarities. This approach confirms our initial hypothesis that evaluating a model's ability to convert notation to music should not be limited to textual accuracy, as in OCR, but should also consider auditory differences, particularly for non-experts' everyday needs. Additionally, we explored the varying outcomes of using different embeddings for FAD scores and their correlation with human perceptions, an aspect previously underemphasized. Given the rapid advancements in multimodal large models, we anticipate the emergence of more models capable of performing this task in the future.

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